ML Task-6

You are working as a data scientist for an email service provider. The company wants to automatically classify incoming SMS messages as spam or ham (not spam).

The dataset contains thousands of messages, and after applying text processing techniques like TF-IDF vectorization, each message is represented by a large number of word-based features. However, many of these features are irrelevant and may negatively affect model performance.

To handle this high-dimensional data and perform feature selection, you decide to use Lasso Regression, which can shrink some feature coefficients to zero and retain only the most important words for classification.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

df = pd.read_csv("spam.csv", encoding='latin-1')

df = df[['v1', 'v2']]
df.columns = ['label', 'message']

df['label'] = df['label'].map({'ham': 0, 'spam': 1})

vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(df['message'])

total_features = X.shape[1]
print("Total number of features:", total_features)

y = df['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

def train_lasso(alpha):
    model = LogisticRegression(penalty='l1', solver='liblinear', C=1/alpha, max_iter=5000)
    model.fit(X_train, y_train)

    coef = model.coef_[0]
```

```
    non_zero = np.sum(coef != 0)
    zero = np.sum(coef == 0)

    print("\nAlpha =", alpha)
    print("Non-zero coefficients:", non_zero)
    print("Eliminated features:", zero)

    return non_zero

features_001 = train_lasso(0.01)
features_01 = train_lasso(0.1)
features_1 = train_lasso(1)

reduction = ((total_features - features_01) / total_features) * 100
print("\nFeature Reduction Percentage (alpha=0.1):", reduction)

alphas = [0.01, 0.1, 1]
features = [features_001, features_01, features_1]

plt.figure()
plt.plot(alphas, features, marker='o')
plt.xlabel("Alpha")
plt.ylabel("Number of Selected Features")
plt.title("Alpha vs Selected Features")
plt.show()
```

OUTPUT :

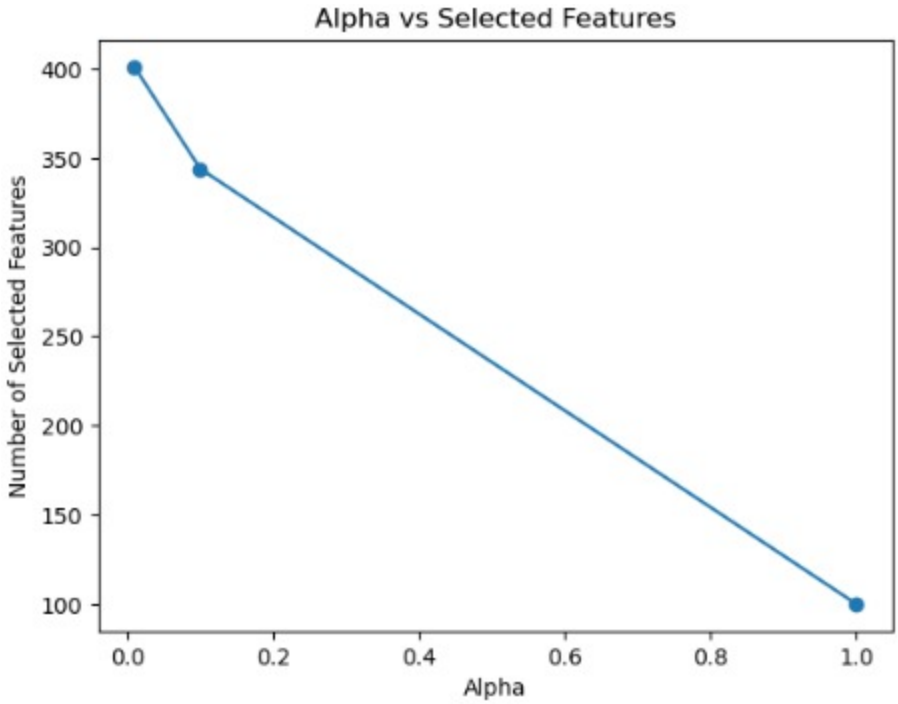
```
Total number of features: 8404

Alpha = 0.01
Non-zero coefficients: 401
Eliminated features: 8003
D:\Anaconda3\Lib\site-packages\sklearn\svm\_base.py:1249: Convergen
  warnings.warn(

Alpha = 0.1
Non-zero coefficients: 344
Eliminated features: 8060

Alpha = 1
Non-zero coefficients: 100
Eliminated features: 8304

Feature Reduction Percentage (alpha=0.1): 95.90671108995717
```


Alpha vs Selected Features
Number of Selected Features
400
350
300
250
200
150
100
0.0
0.2
0.4
0.6
0.8
1.0
Alpha